



EAST WEST UNIVERSITY
Department of Mathematics & Data Science
Assignment, Summer 2026 Semester

Course:	PHY 209 : Engineering Physics – II
Instructor:	Md. Mahfuzul Haque, Assistant Professor, Department of MDS

1. A particle limited to the x axis has the wave function $\Psi = ax$ between $x = 0$ and $x = 1$; $\Psi = 0$ elsewhere. (a) Find the value of a , (b) Find the probability that the particle can be found between $x = 0.45$ and $x = 0.55$. (b) Find the expectation value x of the particle's position.
2. A particle is confined to the x -axis and has the wave function: $\Psi(x) = a(1-x)$ for $0 \leq x \leq 1$; $\Psi(x)=0$ elsewhere. (a) Find the probability that the particle can be found between $x = 0.3$ and $x = 0.6$. (b) Find the expectation value x of the particle's position.
3. The wave function of a certain particle is $\Psi(x) = A \sin 2x$ for $-\pi/2 < x < \pi/2$. Find the probability that the particle be found between $x = -\pi/4$ and $x = \pi/4$.
4. An eigenfunction of the operator d^2/dx^2 is $\Psi = \sin nx$, where $n = 1, 2, 3, \dots$. Find the corresponding eigenvalues.
5. Consider an wave function: $\psi(x, t) = Ae^{i(kx - \omega t)}$. Determine the momentum eigenvalue and the energy eigenvalue corresponding to this wave function.
6. Find the probability that a particle trapped in a box L wide can be found between $0.45L$ and $0.55L$ for the ground and first excited states.
7. Find the expectation value $\langle x \rangle$ of the position of a particle trapped in a box L wide.
8. Consider two states

$$|\psi_1\rangle = (1 + i)|\phi_1\rangle - 2|\phi_2\rangle + b|\phi_3\rangle + 3i|\phi_4\rangle$$

and

$$|\psi_2\rangle = 2|\phi_1\rangle + i|\phi_2\rangle - 4|\phi_3\rangle + |\phi_4\rangle$$

where $|\phi_1\rangle, |\phi_2\rangle, |\phi_3\rangle$, and $|\phi_4\rangle$ are orthonormal kets, and where b is a constant. Find the value of b such that $|\psi_1\rangle$ and $|\psi_2\rangle$ are orthogonal.

9.

Consider the states $|\psi\rangle = 3i|\phi_1\rangle - 7i|\phi_2\rangle$ and $|\chi\rangle = -|\phi_1\rangle + 2i|\phi_2\rangle$, where $|\phi_1\rangle$ and $|\phi_2\rangle$ are orthonormal.

- Calculate $|\psi + \chi\rangle$ and $\langle\psi + \chi|$.
- Calculate the scalar products $\langle\psi|\chi\rangle$ and $\langle\chi|\psi\rangle$. Are they equal?
- Show that the states $|\psi\rangle$ and $|\chi\rangle$ satisfy the Schwarz inequality.
- Show that the states $|\psi\rangle$ and $|\chi\rangle$ satisfy the triangle inequality.

10.

$$|\psi\rangle = \begin{pmatrix} -3i \\ 2+i \\ 4 \end{pmatrix}, \quad |\phi\rangle = \begin{pmatrix} 2 \\ -i \\ 2-3i \end{pmatrix}.$$

- Find the bra $\langle\phi|$.
- Evaluate the scalar product $\langle\phi|\psi\rangle$.
- Examine why the products $|\psi\rangle|\phi\rangle$ and $\langle\phi|\langle\psi|$ do not make sense.

11. Show that linear momentum operator $\hat{p}_x$ and the Hamiltonian operator $\hat{H}$ commute each other, when the potential energy V is constant.

12. Show that

$$\left[\hat{x}, \frac{\partial}{\partial x} \right] = -1.$$

13. Show that

$$\hat{x}^2 \frac{d}{dx} \frac{1}{\hat{x}} = \hat{x} \frac{d}{dx} - 1.$$

14. Show that

$$[\hat{p}_x, \hat{V}(x)] = -i\hbar \frac{dV(x)}{dx}.$$

15. Consider the following two kets:

$$|\alpha\rangle = \begin{pmatrix} 2i \\ -1 \\ 3 \end{pmatrix}, \quad |\beta\rangle = \begin{pmatrix} 4 \\ -2i \\ i \end{pmatrix}$$

Determine whether the kets $|\alpha\rangle$ and $|\beta\rangle$ are orthogonal. If they are not orthogonal, suggest an appropriate change in one component of either ket such that the two kets become orthogonal.